

Single-Channel Fetal QRS Detection: Where the Performance Actually Comes From

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Abstract

Deep detectors for fetal QRS in single-channel abdominal ECG are reported as a whole, so it is unclear which design choices produce the accuracy. We hold the protocol fixed (leave-one-record-out, ± 50 ms, threshold from an inner validation record, label-blind lead selection), vary one factor at a time, and estimate every effect per woman. A dilated temporal convolutional network with 113 481 parameters reaches a macro F1 of 99.21% (95% CI 97.9–100.0) on ADFECGDB and 93.30% (84.9–99.4) zero-shot on ten pregnancy recordings of the Silesian database. A three-tier decomposition attributes the gain to the front-end (the 10–60 Hz band of Xu et al., +11.00 F1, on a 300 ms window learner), to context and per-sample output (+4.49, CI 1.6–7.4) and to the architecture family (+0.41, not resolvable at $n = 5$). Repeated on the TCN itself over 22 women the same band change is worth only +2.44 (CI [−0.04; 6.29]), three women carrying all of it, so tier 1 does not transfer; that run is preliminary. Re-run by us under GNU Octave on all 22 women with the same matcher, the Power-MF pipeline gains 12.12 F1 points from its four channels over its own one-channel restriction (86.71 to 98.83); the network recovers 10.85 of those points (89.5%) from one lead, so it is not separable from four-channel Power-MF (−1.27 F1, CI [−3.08; +0.27], $p = 0.16$, better in 18 of 22 women, median +0.23) and is ahead of one-channel Power-MF (+10.85, 22 of 22). CinC 2013 set-a is not independent of ADFECGDB, as its organisers documented; we identify by measurement the 15 set-a records that are verbatim copies of our training recordings and score the remaining 60: training on 22 women instead of five raises the label-blind score from 64.04 to 74.28 ($p = 3 \cdot 10^{-10}$), and a lead rule driven by the detector’s own output reaches 82.01, a post-hoc result reported as a hypothesis. Short-term variability inherits detection error one-sidedly (bias +20.50 ms on CinC 2013) and is not a standalone variability monitor.

1. Introduction

Fetal heart rate from abdominal electrodes needs a reliable fetal QRS (fQRS) detector; a single-channel one keeps the hardware small. The PhysioNet/CinC Challenge 2013 fixed the metric still in use, F1 with a ± 50 ms window [1]; Behar et al. benchmarked template subtraction, Kalman and blind-source methods under it, grid-searching the band-pass filter [2]; Andreotti et al. showed on synthetic signals that pre-processing changes the ranking of extraction algorithms [3]. Deep single-channel methods since [4] change many things at once: dual-path source separation [5], a matched-filter pipeline with power-spectral-density (PSD) lead selection over four channels [6], and guided empirical mode decomposition with a 10–60 Hz band [7].

Several ingredients of our detector are established elsewhere and we claim none as new: per-sample R-peak output instead of window classification [8], dilation rather than kernel size as the useful knob in fECG [9], and the band itself [7]. Each of those papers changes band, sampling rate, context, architecture and sometimes the lead rule at once, and the lead is often chosen with the test labels; what is missing is how much each choice is worth once the protocol is held fixed. The contributions are:

- a leakage-free single-channel protocol with a three-tier decomposition of the gain into front-end, temporal context and architecture family, analysed per woman;
- a like-for-like comparison against a multi-channel published pipeline re-run by us and scored with the same matcher, rather than quoted from its paper;
- cross-dataset evidence on 22 independent women and on the 60 CinC 2013 set-a records that are not copies of the training data, that generalisation breaks across recorders rather than gestation, and the label-blind lead rule with

it.

2. Methods

2.1. Data

ADFECGDB [10]: five labour recordings (r01, r04, r07, r08, r10), four abdominal leads at 1 kHz, 5 min each, with fQRS annotations from a scalp electrode (3 191 beats). **Silesian database** [11]: ten pregnancy recordings (B1, 32–42 weeks, 199.6 min total) and twelve labour recordings (B2, 5 min), four abdominal leads at 500 Hz, same hospital and recorder as ADFECGDB. B1 annotations are indirect (author pipeline plus expert correction) and in 5/10 records sit 8–12 ms after the abdominal R peak; B2 annotations come from a scalp electrode. The overlap with ADFECGDB was noted by Baldazzi and Pani [12]; we cross-correlated all 4×4 lead pairs at 250 Hz and found five B2 records that are ADFECGDB records (B2_01 = r01, B2_02 = r10, B2_07 = r04, B2_10 = r07, B2_11 = r08; correlation 0.988–0.994). The independent set is therefore 22 women (5 ADFECGDB, 10 B1, 7 new B2); duplicates are scored only with the fold model that never saw them. **CinC 2013 set-a** [13, 1]: 75 records, four channels at 1 kHz, 1 min each, a different recorder. Set-a is *not* independent of ADFECGDB: as the organisers documented, 25 of the 447 Challenge records were derived from it [13, 1], and Clifford et al. cautioned that an entry trained on ADFECGDB “may have led to a bias in the results as this database was included in set-a, set-b (and possibly a few records in set-c)” [1]. No accessible source states *which* set-a records they are; by normalised cross-correlation of the raw signals over all lags (two independent implementations) exactly 15 — a03–a05, a08, a12–a15, a17, a19, a20, a22–a25 — are verbatim copies (NCC = 1.0000 on all four channels, $|\Delta RR| = 0.0$ ms) of the 0–60, 120–180 and 240–300 s windows of r01, r04, r07, r08 and r10; the other 60 reach at most 0.62. Every CinC figure below uses those 60 records.

2.2. Detector

One abdominal lead is band-passed 10–60 Hz (4th-order Butterworth, zero-phase) with a 50 Hz notch and resampled to 250 Hz; the band is taken from [7] and Sec. 3.4 measures what it is worth against seven other published bands. Maternal QRS are located on an 8–25 Hz envelope and removed by subtracting a median beat template with per-beat least-squares scaling. The residual and the filtered signal, each robust-scaled, form a two-channel 4 s input. The network is a dilated residual TCN: a stem and five residual blocks (dilations 1, 2, 4, 8, 16; 40 channels; kernel 7; BatchNorm; GELU) with a 1×1 output convolution, 113 481 parameters, receptive field 1 516 ms. It emits one logit per sample against a Gaussian heat-map target ($\sigma = 3$ samples) at each annotated fQRS [8], trained with weighted binary cross-entropy and AdamW (3×10^{-3} , one-cycle), six epochs, batch 32, 1 s stride. At inference, overlapping windows are averaged and peaks above a threshold with a 250 ms refractory period are the fQRS (0.48 MB, 4.35 ms per window on a desktop CPU).

2.3. Protocol and statistics

Leave-one-record-out (LORO) on ADFECGDB with five folds: one test, one inner validation and three training records; the threshold is chosen on the validation record from a grid 0.20–0.75 and applied unchanged. Detections are matched one-to-one to annotations by greedy nearest matching within ± 50 ms. Because a device sees one lead, we also report a *label-blind* lead: following Power-MF [6], the lead whose residual envelope has the largest Welch PSD peak in 1.8–3.0 Hz, a rule that never touches annotations. Silesian and CinC records are scored zero-shot. A second model, used only where stated, is trained on all 22 independent women with 11-fold grouped cross-validation.

The unit of analysis is the woman, not the (record $\times$ lead) pair: four leads of one recording are repeated measures in one cluster, and treating them as 20 observations inflates every test. We average leads within a woman and report mean differences with 95% cluster-bootstrap percentile intervals over subjects (10 000 resamples, seed 0); where these disagree with the paired- t interval at $n \leq 7$ we give both and trust the latter. “No difference” claims use two one-sided tests (TOST) at a pre-declared 1.0-point margin. With $n = 5$ the smallest two-sided p an exact signed-rank test can give is 0.0625, so no ADFECGDB comparison can reach $p < 0.05$ at subject level; we quote intervals there, not p -values.

2.4. Experiments

Baselines, re-implemented and scored with the same matcher: template subtraction (TS) and TS with a two-component principal-component template (TS-PCA) [2], each with an adaptive Pan–Tompkins detector [14], and *Prominence*, our front-end followed by *scipy* peak picking; hyper-parameters were tuned on r01 only. **Power-MF** [6] was not quoted but executed: the authors’ MATLAB pipeline was run under GNU Octave 11.3 and scored with *our* matcher at ± 50 ms, at four leads with both ICA stages and again restricted to the single PSD-selected lead, so all three columns share a scoring rule.

Three-tier decomposition. Tier 1 (front-end): eight literature band-pass settings, LORO, gradient-boosted trees on 300 ms windows of two leads. Tier 2: a 300 ms window classifier versus the 4 s per-sample TCN. Tier 3 (family at fixed

Table 1: Macro F1 (%) of one model trained on five ADFECGDB women, ± 50 ms; CI is a 95% cluster bootstrap over records, “ ≥ 90 ”/“ < 50 ” count PSD-lead F1 in that range.

Configuration	n	PSD lead	95% CI	4-lead mean	≥ 90	< 50
ADFECDGB, LORO	5	99.21	[97.9; 100.0]	97.45	5	0
Silesia B2 labour, all	12	97.17	[92.7; 99.8]	95.26	11	0
7 new women, zero-shot	7	95.87	[88.4; 99.8]	93.73	6	0
Silesia B1 pregnancy, zero-shot	10	93.30	[84.9; 99.4]	91.31	8	0
CinC 2013 set-a, 60 non-ADFECDGB records	60	64.04	[55.4; 72.5]	56.00	27	22

context): gradient boosting versus a dilated CNN on the same windows. A receptive-field sweep (172–3 052 ms) isolates context within one family, and an architecture benchmark compares eight families on 300 ms windows.

Reject gate. A 4 s segment is *bad* if the detector’s F1 on it is below 80. A classifier trained on 1 500 ADFECGDB segments and tested on 600 CinC 2013 segments uses 16 topological [15], 12 classical [16], or all features; intervals resample records, not segments.

3. Results

3.1. Detection across data configurations

Table 1 gives the same model under the same protocol on five configurations. On ADFECGDB the label-blind PSD rule reaches 99.21%, identical to the oracle lead in every record, and 99.18% on the DPSS benchmark leads [5]; across four leads the LORO macro F1 is 97.45%, 97.16% over three seeds. Zero-shot on Silesian pregnancy records it keeps 93.30%, eight of ten records above 90 (oracle lead 97.04%); the two weakest are those where the PSD rule picks the worst lead. A second seed of the 22-subject model moves per-subject F1 by 0.28 points on average (largest 2.81), though per-fold thresholds move more.

3.2. How much of multi-channel separation one channel recovers

Competing numbers in the previous draft were quoted, not measured, so we ran Power-MF [6] under Octave and scored it with our matcher. An earlier run covered only 16 of 22 women because the port returned nothing, or half the beat rate, on six long B1 recordings; *those figures are withdrawn*. The fault was ours: Octave’s *findpeaks* enforces *MinPeakDistance* with an $O(k^2)$ distance matrix that overflows on B1 (2 395 600 samples after $\times 4$ interpolation), whereas MATLAB resolves it greedily. We re-implemented the MATLAB semantics in $O(k)$ (48/48 test cases); the rival explanation, a 340 ms minimum peak distance below the fetal RR interval, was rejected (0.00% of B1_01’s RR intervals are shorter). The repaired port scores 99.40 ± 0.51 on the ten B1 records against 99.46 published by the authors. All 22 women are scored, with no exclusion (Table 2).

This makes the question explicit: how much of what *multi-channel source separation* buys can a 113 k single-channel network recover? Restricting Power-MF to one lead costs it 12.12 F1 points (98.83 to 86.71, CI $[-18.27; -6.90]$, worse in 21 of 22 women), which is the price of the four-channel ICA stages on this material. Our single-channel network recovers 10.85 of those points ($+10.85$, CI $[+6.80; +15.40]$, $p = 5 \cdot 10^{-7}$, better in 22 of 22 women), that is 89.5% of the benefit, from one electrode pair. Against the full four-channel pipeline it is not separable (-1.27 , CI $[-3.08; +0.27]$, $p = 0.16$), and on ADFECGDB the interval excludes zero in its favour ($+0.39$, CI $[+0.22; +0.60]$, 5/5 women).

Table 2: Power-MF [6] *re-run* by us, at four channels and restricted to one lead, against our 22-subject single-channel model: same recordings, same ± 50 ms matcher. Δ_4 and Δ_1 are ours – Power-MF per woman with 95% cluster-bootstrap intervals (10 000 resamples of subjects; of records for CinC). Power-MF was not run at four channels on CinC 2013.

Subject set	<i>n</i>	mean F1 (%)			ours – Power-MF	
		PMF-4	PMF-1	ours	Δ_4 [95% CI]	Δ_1 [95% CI]
ADFECDGB	5	99.01	92.37	99.40	+0.39 [+0.22; +0.60]	+7.02 [+3.84; +9.34]
Silesia B2, new	7	97.90	87.28	96.82	−1.08 [−4.04; +0.54]	+9.54 [+3.09; +17.9]
Silesia B1	10	99.40	83.48	97.15	−2.25 [−5.55; +0.32]	+13.68 [+6.91; +21.3]
All 22	22	98.83	86.71	97.56	−1.27 [−3.08; +0.27]	+10.85 [+6.80; +15.4]
CinC 2013, 60 clean records	60	—	55.97	74.28	—	+18.32 [+13.4; +23.6]

The mean difference against four channels is negative, but it is not a uniform deficit: the median difference over the 22 women is +0.23 and the network is better in 18 of them. The mean is pulled below zero by three recordings — B1_07 (−12.88), B1_06 (−10.51) and B2_03 (−9.77) — and mildly by B1_10 (−1.86). The defensible statement is that one channel recovers most of the multi-channel benefit, not that it beats four channels.

3.3. CinC 2013: the 60 records that are not training data

Every CinC figure in earlier drafts is withdrawn: the first drafts scored a ten-record subsample (59.15 / 69.31 under the blind rule, 77.34 / 90.34 under a lead chosen after the fact), the next scored all 75 records (71.21 / 79.40 / 86.87), and both sets include the 15 records that are copies of the five women every model was trained on. Those 15 records score 99.87 under the 22-woman model and inflated the 75-record means by 3.27–7.18 points. Table 3 reports the five-woman (m5) and 22-woman (m22) models on the 60 remaining records, and on the 53 left after also removing a33, a38, a47, a52, a54, a71 and a74, whose reference annotations are known to be unreliable [2] (that list was declared before the run). Under the label-blind PSD rule, training on 22 women instead of five raises the mean from 64.04 to 74.28 (+10.24, CI [7.13; 13.60], $p = 3.5 \cdot 10^{-10}$, better in 52 records and worse in 3); records at or above 90 go from 27 to 33 and those below 50 from 22 to 16; the 53-record variant gives +11.08 ($p = 4.4 \cdot 10^{-9}$). The median record of m22 is 95.07: the distribution is split, not shifted.

The lead rule loses accuracy across recorders before the detector does. In domain the PSD rule is 0.02–1.90 points below the oracle lead; on the 60 CinC records it is 9.32 below (74.28 vs 83.60, CI [5.10; 14.15]) and picks the oracle lead in 17 of 60. Seven label-blind alternatives were compared on the same records under a written analysis plan that designated a gate-score rule as confirmatory; that rule gains +6.44 but fails Holm correction ($p_{\text{Holm}} = 0.051$), so the confirmatory test failed. The best rule, *peakprob* — run the detector on all four leads and keep the lead whose mean probability at its own detected peaks is highest — gains +7.73 (CI [3.82; 12.41], $p_{\text{Holm}} = 0.0039$ over the seven rules, 19 wins / 35 ties / 6 losses) and reaches 82.01, but it was chosen after the data were seen, on the logit scale a different rule (gate4) leads, and no public dataset with real fQRS labels exists on which to replicate it (NIFEADB, NInFEA and set-b ship none). We report it as a hypothesis, not a result. A label-shuffle check changed 0 of 776 lead choices, so the selection itself is label-blind.

Table 3: CinC 2013 set-a without the 15 ADFECGDB copies, zero-shot: mean F1 (%) after training on five women (m5) and on 22 (m22). Δ is per record, 95% bootstrap over records; p exact signed-rank; “ ≥ 90 ”/“ < 50 ” count m22 records. Only the top three rows are label-blind; the *peakprob* row is a post-hoc choice.

Lead rule	<i>n</i>	m5	m22	Δ [95% CI]	<i>p</i>	≥ 90	< 50
PSD, label-blind (declared)	60	64.04	74.28	+10.24 [7.1; 13.6]	$3 \cdot 10^{-10}$	33	16
4-lead mean	60	56.00	67.69	+11.69 [9.0; 14.4]	$3 \cdot 10^{-11}$	15	17
<i>peakprob</i> (post hoc)	60	—	82.01	—	—	38	9
Fixed lead 0 (post hoc, withdrawn)	60	48.43	61.78	+13.35 [9.7; 17.1]	$5 \cdot 10^{-9}$	21	27
Oracle lead (labels)	60	72.76	83.60	+10.84 [7.5; 14.4]	$1 \cdot 10^{-9}$	38	8
PSD, 53 records	53	64.19	75.27	+11.08 [7.5; 14.8]	$4 \cdot 10^{-9}$	31	14

3.4. Where the performance comes from

Three tiers decompose the gain on ADFECGDB (LORO, ± 50 ms; tier 1 on two leads with a gradient-boosted learner on 300 ms windows, tiers 2–3 on four leads). *Tier 1, front-end:* changing only the band from 1–45 Hz to 10–60 Hz moves F1 from 80.06 to 91.06, +11.00 points (not stored per record, so no interval); the spread over eight literature bands is 18.89 points (72.17 for 0.5–100 Hz to 91.06 for 10–60 Hz). Re-run on the TCN itself over the 22 women (three folds, one seed, four epochs) the same change gives 95.63 to 98.07, +2.44, cluster bootstrap [−0.04; 6.29] containing zero: at $n = 22$ the band effect is *not* established on the model we use. Three women (B2_03 +37.8, B1_06 +8.8, B1_07 +8.2) carry all of it, the other 19 average −0.07, and averaging the four leads instead of selecting one leaves −0.07 [−0.51; 0.39] — the gain acts through the lead rule, not the band. *Tier 2, context and output:* a 300 ms window classifier to the 4 s per-sample TCN, 92.67 to 97.16, +4.49 per woman [1.58; 7.43] (the conservative paired-*t* interval [−0.29; 9.27] still contains zero at $n = 5$); this step bundles receptive field, output parameterisation and parameter count (33 k to 113 k). *Tier 3, family:* gradient boosting to a dilated CNN at the same context, 92.49 to 92.90, +0.41, inside the 1.0-point equivalence margin and not resolvable either way. Within the TCN family, LORO F1 rises from 92.67 at a 172 ms receptive field to 97.16 at 1 516 ms, and the validation-record sweep saturates near 1.5 s.

Against re-implemented baselines on ADFECGDB (four-lead F1: TS 78.96, TS-PCA 91.05, Prominence 86.39, ours 97.45) the network gains 11.06 points over peak picking on the residual it receives (CI [7.80; 14.51], 5/5) and halves the spread (SD 4.44 vs 10.17). The 18.49-point margin over TS is unstable (paired-*t* [−1.81; 38.80], one record drives it), and TS-PCA on the PSD lead (96.74) is 2.47 points below.

In the architecture benchmark six families lie within 1.53 points (92.43 to 90.90); TOST at the 1.0-point margin declares 0 of 7 equivalent and 6 of 7 inconclusive, so the absent difference measures the sample size. Two effects clear it: the plain CNN without dilation is worse by 2.95 (90% CI [−4.55; −1.47]), the large CNN 0.73 above ([0.18; 1.22]). All eight, at 300 ms, stay below the 4 s per-sample model.

3.5. Reject gate, topology, clinical read-out

The 12 classical features predict bad segments with AUROC 0.929 (CI [0.830; 0.982]) and lift retained-segment F1 from 61.96 at full coverage to 69.40 at 80% (random 62.03). That pooled figure is largely *between-record*: over the five records containing both classes the *within-record* AUROC is only 0.721 ([0.517; 0.898]), and that is what a monitoring session

sees. Topology contributed nothing: 16 topological features reach 0.566 and adding them lowers AUROC to 0.905.

Clinical read-out. Short-term variability (STV, 3.75 s epochs) computed from the detected beats tracks detection quality: bias +0.33 ms on ADFECGDB and +20.50 ms on CinC 2013; mean absolute STV error is 0.13 ms at $F1 \geq 99.5$ and 25.78 ms at $F1 < 90$, and one-sided (25 of 32 records overestimate), since every missed or spurious beat perturbs an RR interval. On gated records the limits of agreement are $[-0.39; +0.61]$ ms, but no in-domain subject has a reference STV near a management threshold, so the detector is usable for trending on gated records and *not* as a standalone STV monitor.

4. Discussion

The front-end dominates the 300 ms learner, not yet the network. Of the 15.9 points gained over the three tiers, 11.0 come from the pass-band alone on that learner: the quantitative form, on real recordings, of what [3] showed on synthetic ones. On the network the same change is worth 2.44, interval containing zero, and nothing when the four leads are averaged instead of selected. The receptive field is the architectural variable we can resolve: dilation (+2.95 at equal parameters) and a longer field (+4.5 from 172 to 1516 ms) clear the 1.0-point margin [9]; other family differences do not.

One channel recovers most of what four channels buy. Multi-channel separation is worth a directly measured 12.12 F1 points to Power-MF itself, and a 113 k single-channel network recovers 10.85 of them, 89.5%. The residual gap against four channels is concentrated, not diffuse: three of 22 recordings account for it and the median woman is +0.23; the failures a single lead creates are few but severe, which is what a device must gate against.

Generalisation breaks across recorders, and the lead rule with it. The model loses about four points on pregnancy records from the same recorder (97 to 93) and about 35 on a different one (99 to 64 under the blind rule, five-woman model, 60 clean records). Part of that is the selection rule: the PSD rule is within 1.90 points of the oracle in domain and 9.32 below it out of domain. Training on 22 women recovers 10.2 points without closing the recorder gap, and four unsupervised adaptation methods (adaptive notch, self-training, AdaBN, TENT) recover none of it. Whether the residual gap is signal that is absent or signal the model misses could not be decided: the visibility test we built for that purpose has an 18.0% false-negative rate on these records. A rule that lets the detector score the leads should replace the spectral heuristic, but it must be fixed before, not after, the next held-out set is opened.

Limitations. The binding limitation is n : five women on ADFECGDB and 22 in all, one hospital, one recorder, every training woman in labour at 38–41 weeks. No ADFECGDB comparison can reach $p < 0.05$ at subject level, every interval is wide, and the architecture benchmark is underpowered by construction. *Against four-channel Power-MF our mean is 1.27 points lower.* The interval $[-3.08; +0.27]$ contains zero, so neither system can be declared better, but the point estimate is against us (B1_07, B1_06, B2_03 carry the deficit). Power-MF was run at our resampled rate with our matcher, not in the authors' own evaluation. The 10–60 Hz band is not our finding but Xu et al.'s [7]; our contribution is only

the measurement of what it is worth. That measurement (tier 1, +11.00) was made with a gradient-boosted learner on 300 ms windows; on the TCN it falls to +2.44 with an interval containing zero, so tier 1 is not established on the network and +11.00 must be read as bounded to the 300 ms learner. That run is preliminary: one seed. B1 labels are indirect, so B1 F1 measures agreement with the authors' pipeline, and the model trained on B1 sits 3.25 ms from those labels against 6.50 ms for models that never saw them, so B1 must not be used to argue timing accuracy. DPSS [5] numbers remain quoted, not re-run. The main model uses one seed and the CinC figures one in-domain threshold per model. One topological configuration was tried, without tuning on test labels.

5. Conclusion

With protocol held fixed, single-channel fQRS detection on ADFECGDB reaches 99.21% F1 on a label-blind lead and 93.30% zero-shot on pregnancy recordings. Four-channel Power-MF gains 12.12 F1 points over its own single-channel restriction; the network recovers 10.85 of them (89.5%) from one lead, not separable from four channels (-1.27 , CI $[-3.08; +0.27]$) and above one channel (22/22 women). The 10–60 Hz front-end of [7] is worth 11.00 points to a 300 ms learner but only 2.44 to the network; the family is a variable five women cannot resolve. Cross-recorder transfer is the open problem: on the 60 CinC 2013 set-a records that are not copies of the training data the blind score is 74.28% after training on 22 women, and the lead rule degrades before the detector does.

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